

Test 29: Dirac dip in DOS at $\gamma=1/2$

Verdict: PASS | Generated: 2026-09-15 04:28:22 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Semi-exact: DOS dip depth vs prediction.

METHOD: Semi-exact: DOS dip depth vs prediction.

Test 29 SERIES: 9-pt fine scan @96 γ (strict dip true, $\gamma_{\text{dip}}=0.0195$), contrast(w) 288.00 $\pm$ 3.97 ($w=0.1\pm0.8$), size contrast 13.33 $\times$ 10.67 $\times$

Result: [SERIES pass 3]: 3 sub-checks, 0 failed γ PASS

- Plots
- 'plots/plot_01.png' γ 1600 $\times$ 1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_02.png' γ 1600 $\times$ 1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_03.png' γ 1600 $\times$ 1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_04.png' γ 1600 $\times$ 1000, ABPlotV23 engine (supersampled)
 - 'plots/animation.gif' γ animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 29 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[03:49:19.515] [+14550.543s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\gamma=0.3958$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[03:53:34.081] [+14805.109s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\gamma=0.4271$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[03:58:04.145] [+15049.123s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\gamma=0.4583$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:02:28.229] [+15339.257s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\gamma=0.4896$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:06:49.977] [+15601.005s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\gamma=0.5000$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:11:11.976] [+15863.005s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\gamma=0.5104$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:15:30.824] [+16121.852s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\gamma=0.5417$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:19:51.305] [+16382.334s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\gamma=0.5729$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:24:02.022] [+16633.050s] build_ab_cloud_hamiltonian: 96x96 (N=9216), $\gamma=0.6042$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:28:08.106] [+16879.134s] build_ab_cloud_hamiltonian: 48x48 (N=2304), $\gamma=0.5000$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:28:14.173] [+16885.201s] build_ab_cloud_hamiltonian: 48x48 (N=2304), $\gamma=0.3958$, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

CONSOLE CAPTURE

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TEST 29 (pass 3, SERIES): Dirac dip ? fine ?-scan + window series

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Fine scan: 9 points ? = k/96 around k=48 (?=1/2 EXACT), window |E|<0.5

? RUNTIME: 9 solves at 96? + 2 anchors at 48? ? 8?13 min

?=38/96 = 0.39583: ?(|E|<0.5) = 0.2083 (253.5 s)

?=41/96 = 0.42708: ?(|E|<0.5) = 0.1458 (265.2 s)

?=44/96 = 0.45833: ?(|E|<0.5) = 0.0833 (262.8 s)

?=47/96 = 0.48958: ?(|E|<0.5) = 0.0208 (259.8 s)

?=48/96 = 0.50000: ?(|E|<0.5) = 0.0195 (259.3 s)

?=49/96 = 0.51042: ?(|E|<0.5) = 0.0208 (257.2 s)

?=52/96 = 0.54167: ?(|E|<0.5) = 0.0833 (258.4 s)

?=55/96 = 0.57292: ?(|E|<0.5) = 0.1458 (249.0 s)

?=58/96 = 0.60417: ?(|E|<0.5) = 0.2083 (245.2 s)

window |E|<0.1: ?_dip(?=1/2) = 0.0004, ?_off(?=38/96) = 0.1250 ? contrast 288.00×

window |E|<0.2: ?_dip(?=1/2) = 0.0039, ?_off(?=38/96) = 0.2083 ? contrast 53.33×

window |E|<0.3: ?_dip(?=1/2) = 0.0091, ?_off(?=38/96) = 0.2083 ? contrast 22.86×

window |E|<0.5: ?_dip(?=1/2) = 0.0195, ?_off(?=38/96) = 0.2083 ? contrast 10.67×

window |E|<0.8: ?_dip(?=1/2) = 0.0525, ?_off(?=38/96) = 0.2083 ? contrast 3.97×

contrast @w=0.5 (?=0.396 vs 1/2): 13.33×(48x48) ? 10.67×(96x96) ? holds/sharpens ?

??

strict dip @96? (9 pts) PASS ? ?(?=1/2) = 0.0195 below ALL 8 other scan points

contrast grows as w ? 0 PASS ? contrast(w) 288.00×(w=0.1) vs 10.67×(w=0.5): Dirac ? ? E vs metallic ? ? const

no wash-out 48??96? PASS ? 13.33×? 10.67×at the same ? pair (0.396 vs 1/2; tolerance 0.8, as 29b)

Test 29c [SERIES pass 3]: 3 sub-checks, 0 failed ? PASS